

≡ SUPERFAST **PHARMACOLOGY** **≡ REVISION ≡**

• * **HOURS SAVING NOTES** * •

• HANDWRITTEN NOTES •

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FOLLOW YOUR OPD GUIDE



Pharmacology

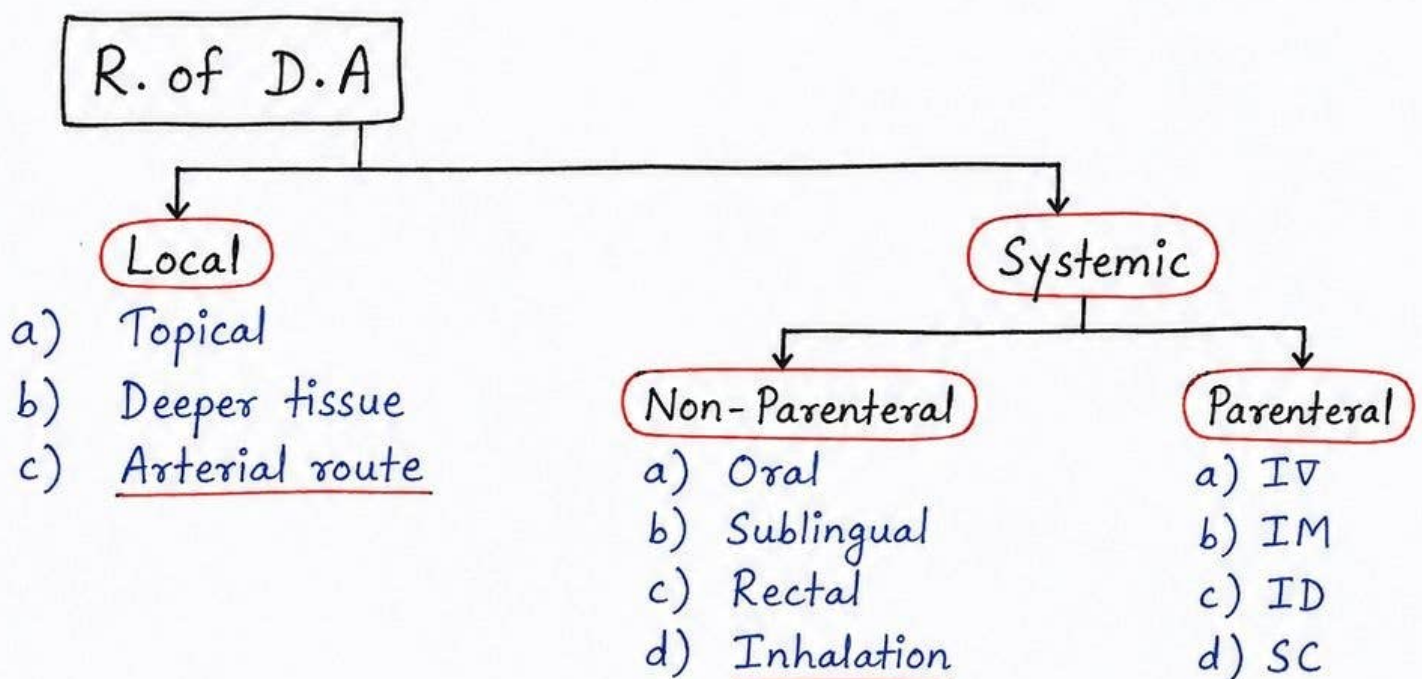
Your **OPD**
Guide

Drug definition → Drogue' - dry herb.

Drug is defined as pharmaceutical drug is also referred to as medicinal product, medicine, medication or drug used to diagnose, cure, treat or to prevent disease.

Drug dosage form → The manner in which drug sub. are presented in the market.

- Type** →
- ① Solid - Tablets, powders
 - ② Liquid - Emulsion, syrup, suspension.
 - ③ Semisolid - Ointment, cream, gels
 - ④ Gaseous - Aerosols, inhalation sprays



* Topical / **Local Route**

- Drug is not absorbed systemically.
- Used for localized action at accessible site.
- Systemic toxicity is absent.

Pharmacology

Your **OPD**
Guide!

(a) Topical

- Refers to external application of drug to the surface for localized action.
- Lotion :- medicine to be rubbed on part.
- Ointment :- medicine to be gently applied.
- Suppositories :- Tablet to be kept in rectum.
- Pessaries :- Vagina
- Lozenges :- Chewable form of tablets for mucous membrane soothing action.

(b) Deeper tissue

- Certain deep areas like knee joint can be approached by using a syringe & needle.
- Inj. that are given in the joint → Intraarticular
e.g. Inj.: Hydrocortisone

(c) Arterial route

- Close intra arterial injection is used for contrast media in angiography.
- Anti cancer drug can be infused in femoral or brachial artery to localize the effect of malignancies.

Systemic route:

- Drugs absorbed into blood stream & distributed all over the body at the site of action through blood circulation.

Non - Parental Route

Drug enter into blood through mucosa.

i) Oral Route (oral mucosa)

- Commonest
- Simplest & oldest

Adv. → Safe, painless, cheap, convenient.

Disadv. → Not useful in unconscious pt.
→ Certain drugs cannot be absorbed orally.
e.g → Insulin, Penicillin.
→ may cause nausea, vomiting.

ii) Sublingual (Buccal mucosa)

- Drug is kept under tongue & absorbed through buccal mucosa.
- Only lipid soluble, non irritating drugs.

Adv. → Rapid absorption directly into systemic circulation.

Disadv. → Small quantity can be given.
e.g → Sorbitrate in angina.
(only lipid soluble drugs.)

iii) Rectal (intestinal mucosa)

- Certain irritant & unpleasant drugs can be put into rectum.

Adv. → This route is convenient when pt. having recurrent vomiting or pt. is unconscious.
e.g → Suppol → highly suitable children.

Disadv. → Absorption is slower & irregular.
→ Inconvenient for pt. & embarassing.
→ Rectal inflammation can result from irritant drugs.

(iv) Inhalation (Nasal mucosa)

Drugs are inhaled through nose or oral cavity which reach deeper mucosa & absorbed into systemic circulation.

Adv. → Action is rapid.

Disadv. → Drug directly reaches to the lungs & heart.

e.g.: Sabutamol in asthma.

(*) Parenteral Route

→ No mucosal barrier.

→ Administration of drug by injection directly into the blood/tissue fluid.

Adv. → 1) Drug action is faster & sure.
2) Valuable in emergencies.

Disadv. → 1) Painful.
→ 2) Require assistance.
→ 3) Possibility of injury to nerve/local tissue.

(i) IM (Intramuscular)

→ Drug injected in large skeletal muscle.

→ e.g. → Gluteus maximus (Rt. upper lateral quadrant)
& Vastus lateralis in pediatric age group.

Selection of Site

- ① Gluteal region — Oil base injection.
- ② Vastus lateralis — Children.
- ③ Deltoid region — non-oily, non-painful inj.
- ④ Triceps muscle — avoided for fear of injury to radial n.

(ii) IV (Intravenous)

→ Drug reaches directly into blood stream & effects are produced immediately.

Adv. → 100% bioavailability

→ Used in emergencies & unconscious pt.

Disadv. → Thrombophlebitis (Inflamm° of veins).

(iii) Subcutaneous

→ Drug is deposited in loose subcutaneous tissue (beneath the skin) which is richly supplied by the but less vascular.

e.g → Inj. of Insulin.

(iv) Intradermal

→ Drugs are injected into the skin raising a bleb.

e.g → BCG vaccine.

(*) Parenteral Route

→ Drug enter into blood stream & effect are produced without passing through G.I.T.

Pharmacokinetics ['K B D T']

- What the body does to the drug.
- Quantitative study of drug movement.
- Determines the route of ad., dose, latency of onset, time of peak action, duration of action & frequency of ad. of drug.

4 Processes →
[ADME] 1) Absorption
 2) Distribution
 3) Metabolism
 4) Excretion

① Absorption

- It is the movement of the drug from its site of administration into the circulation.

* Factors Affecting absorption →

- Drug given in watery sol' absorb faster than solid form.
- Larger the adsorbing surface, faster the absorb'n?
- More is the vascularity, more is the absorb'?

② Distribution

- Once the drug absorbed in blood stream, it gets distributed to other tissues.

$$\text{Volume of distr'n} = \frac{\text{Dose administered}}{\text{Plasma conc'n}}$$

③ Blood Brain Barrier

- Only lipid soluble drugs can cross BBB.

③ Biotransformation (metabolism)

→ Chemical determation of drug in the body.

→ Primary site → Liver

→ Metabolism of drug may lead to following-

① Active metabolites become inactive i.e. Inactive
e.g → Ibuprofen to pcm

② Active metabolites from active drug.
e.g → Digitoxin to Digoxin.

③ Activation of inactive drug.
e.g → Levodopa to Dopamine.

④ Excretion

It is the passage out of systemically absorbed drug.

① Urine

④ Sweat

② Faeces

⑤ Saliva

③ Exhaled air

⑥ Milk.

1st order kinetics

Rate of elimination $\propto$ drug conc.

Zero order kinetics / linear kinetics

Rate of elimⁿ is constant.

Pharmacodynamics [DDB]

- What the drug does to the body.
ie (Study of effects of drug on the body.)
- For most drugs it is necessary to know site of action & mechanism of action at the level of organ.
- Factors → Age, sex, wt, genetic factors, diet of pt.

Drug dose

- Study of drug dosage ⇒ Posology
- Plasma half life of drugs decides the exact dose & dosage schedule.
- Wt. of pt. is main influencing factor.
- Plasma half life of a drug is the time taken for its plasma concⁿ to be reduced to half of its original value.

* Principal of Drug action / Interactions

Drug do not impart new functions to any system, organ or cell, they only alter the pace of ongoing activity.

Types → ① Stimulation

Refers to selective enhancement of the level of activity of specialized cells.

e.g → Adrenaline stimulates heart.

② Replacement

Refers to use of natural metabolites, hormones on their congeners in deficiency state.

e.g → Insulin in DM.

③ Depression

Selective diminution of activity of specialized cells &

e.g → Barbiturates Depress CNS.

④ Irritation

Non selective after noxious effect & particularly applied to less specialised cells.

e.g → Bitter ↑ salivⁿ & gastric secretions.

⑤ Cytotoxic action

Selective cytotoxic action for invading parasite or cancer cells attenuating them without significantly affecting the host cell.

e.g → Penicillin.

Bioavailability → It is the measure of fraction of ad. dose of drug that reaches the systemic circulⁿ in the unchanged form.

→ B.A of IV route is 100% but that of oral route

⊗ is low because, —

① Drug may be incompletely absorbed.

③ Absorbed drug may undergo first pass metabolism in liver.

Mechanism of

Drug
Action.

→ Majority of drugs produce their effect by interacting with specific target (usually a protein) like receptor, enzyme, ion channels, transporters.

→ The fundamental mechanism of drug action can be described ---

- 1) By its physical properties
- 2) By chemical interactions
- 3) By altering metabolic processes
- 4) On the basis of receptors.
- 5) On the basis of enzymes & pumps
- 6) On the basis of ion-channels.
- 7) On the basis of transporters.

① By its (drug) physical property → The drug action may occur due to its physical properties like -

- a) Osmotic property → e.g. Osmotic diuretics such as mannitol.
- b) Adsorption → e.g. activated charcoal.

② By chemical interⁿ → The drug action may result from chemical ~~etc~~ reaction.
e.g → Neutralizⁿ of gastric HCl by antacids

③ By altering metabolic processes.

e.g → Sulfonamides interfere with folic acid synthesis in bacteria.

④ Action through receptors → most drug bind with specific receptors to perform their action.

⑤ Action through enzymes & pumps.

① Aspirin inhibits COX (cyclo-oxygenase)

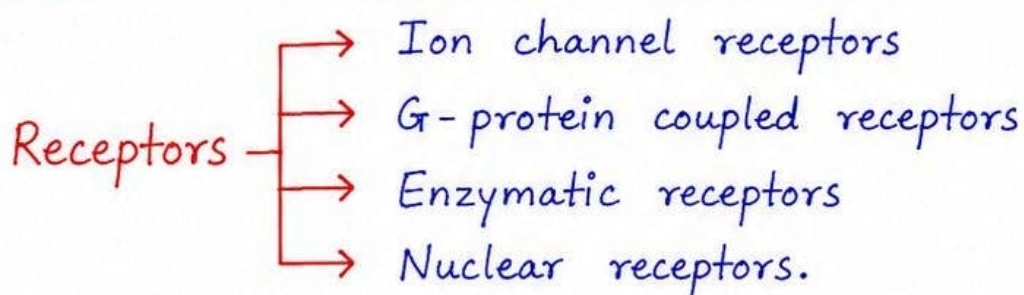
② Omeprazole inhibits H^+ , K^+ ATPase pump.

⑥ Action through ion channels.

e.g → CCB as nifedipine
potassium channel opener as nicorandil.

⑦ Action through transporters.

e.g → Furosemide inhibits Na^+ , K^+ , Cl^- co-transporter in ascending limb of loop of Henle.



DRUGS ACTION ON CNS

A) Anaesthetics

→ Drugs which produce reversible loss of all sensation & consciousness.

→ Cardinal features → 1) Loss of all Sensation (Pain)
2) Sleep
3) Immobility & muscle relaxⁿ.
4) Abolition of somatic & autonomic reflexes.

(MOA) → All anaesthetics cross BBB & inhibit thalamus, RAS & hippocampus by opening chloride channels. These chlorides inhibit & depress the brain so there is loss of consciousness.

Classifⁿ

Given by

(Comp.I)

Inhalation route

(Comp.II)

IV route

Gaseous state

- Nitrous oxide
- Entonox
- Xenon gas

Volatile state

- Ether
- Halothane
- Chloroform

- Thiopental sodium
- Ketamine
- Propofol

Chemical classification

Anaesthetics

Benzodiazepines

- i) Diazepam
- ii) Lorazepam
- iii) Midazolam

Dissociative

- Ketamine

Neuroleptic

- Fentanyl

2 Hypnotics — Sedatives

Sedative → A drug that reduces excitement & calms the subject without inducing sleep, though drowsiness may be produced.
(Anxiolytics)

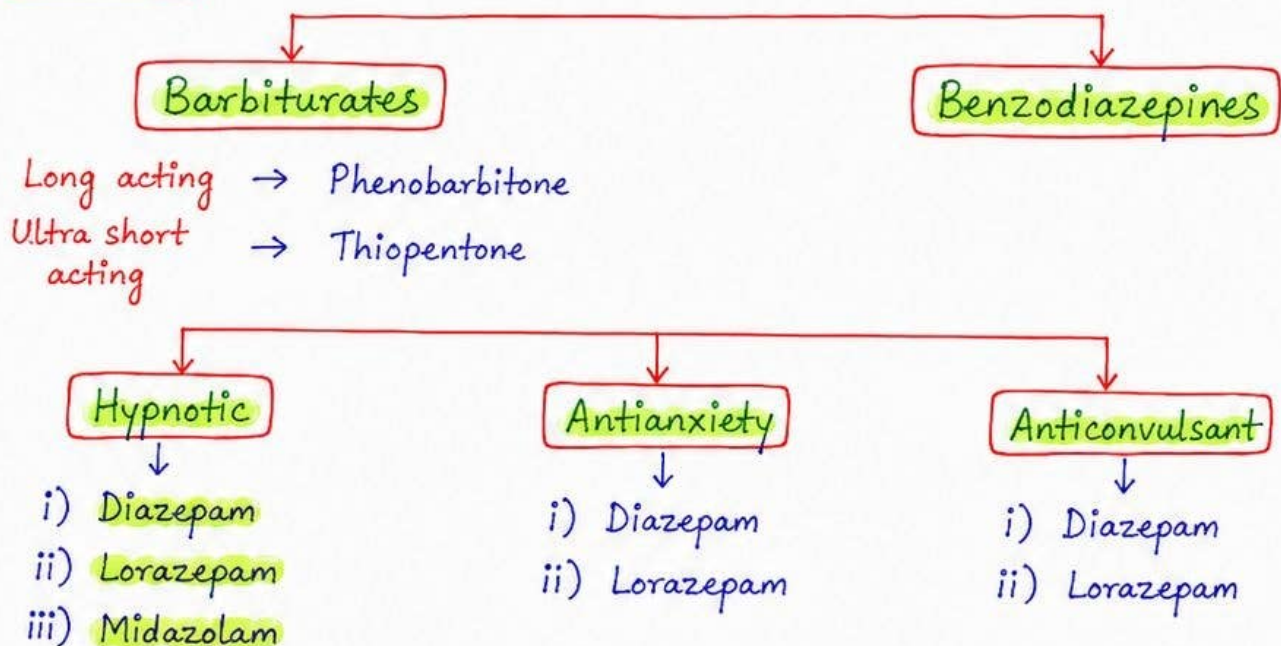
Hypnotic → A drug that induces or maintains sleep, similar to normal arousable sleep.

MOA → Action : midbrain, ascending reticular system & limbic system.

Muscle relaxation is produced by primary action on medulla & ataxia is due to action on cerebellum.

→ These drugs inhibits CNS by opening chloride channels through GABA receptors.

Classify



3 Antiepileptic drugs.

Epilepsy → These are group of disorders of CNS characterized by paroxysmal cerebral dysrhythmia manifesting as brief episodes (seizures) of loss or disturbance of consciousness with or without characteristic body movements (convulsions).

→ Relative deficiency of inhibitory neurotransmitter **GABA (Gamma Amino Butyric Acid)**

MOA → 1) They depress the CNS by stimulating inhibitory receptor GABA.

2) By blocking Na^+ channels.

Antiepileptic drugs are :

1) Phenobarbitone (Gardenal)

2) Phenytoin sodium (Eptoin) ⇒ most popular

3) Carbamazepine (Tegretol)

4) Sodium valproate

↳ D.O.C in Absence seizures especially

5) Levetiracetam ⇒ less sedation

4 Anti-parkinsonism Drugs.

Parkinsonism → Progressive degenerative disorder mostly affecting older people, characterized by rigidity, tremor & hypokinesia.

→ Occurs due to neurotransmitter imbalance, i.e. decrease in conc. of dopamine in basal ganglion.

MOA → Act by promoting pre-synaptic synthesis & release of Dopamine in brain.

Universal antiparkinsonism drug

- Drugs →
- ① Dopamine precursor — Levodopa
 - ② Bromocriptin
 - ③ Selegiline
 - ④ Benzhexol HCl → used in drug induced parkinsonism.

⑤ Anti-depressant drugs

Depression: Is a nervous personality with sadness, loss of interest & pleasure, physical & mental slowing with excessive sleep.

MOA →

- 1) Selectively inhibit monoamine oxidase
- 2) Act as alpha adrenergic blockers.

Drugs → (A) Tricyclic antidepressants

- (ii) Imipramine
- (ii) Amitriptyline

(B) Selective serotonin reuptake inhibitor

- 1) Fluoxetine
- 2) Sertraline
- 3) Citalopram
- 4) Escitalopram

⑥ Anti-anxiety drugs

Anxiety → condition where intense, excessive, persistent worry & fear about every situation.

→ ↑ HR, RR, sweating & tired feeling.

- MOA →
- 1) Benzodiazepines
 - 2) Sedative & hypnotics
 - 3) Beta blockers
 - 4) Selective serotonin reuptake inhibitor.

7 Topical & Analgesics drugs.

- (MOA)
- 1) By ↓ intracellular calcium & release of neurotransmitter & by increasing pain threshold.
 - 2) Depress CNS & produce sleep.

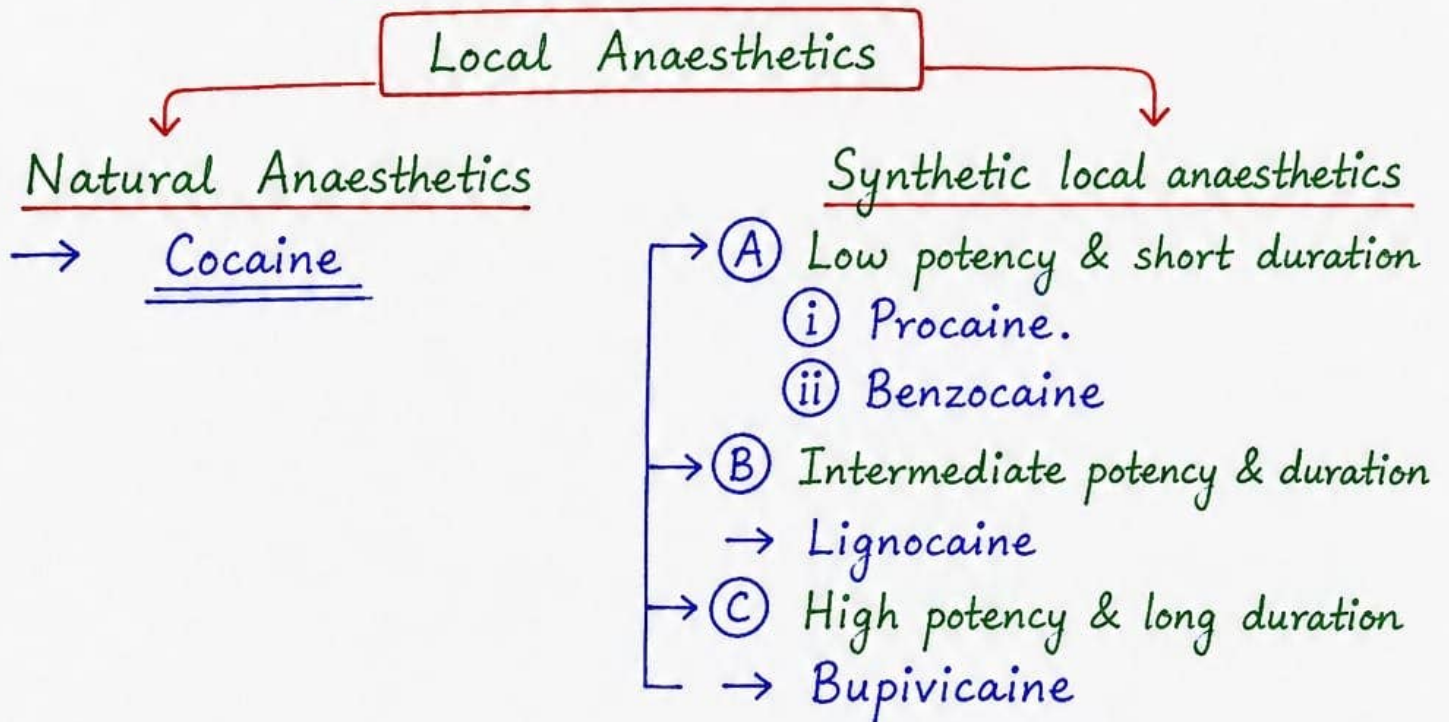
- Drugs →
- [MTP] {
- 1) Morphine
 - 2) Pethidine
 - 3) Pentazocine
 - 4) Tramadol
 - 5) Fentanyl

② Drugs Acting on PNS

→ Local anaesthetics: & Skeletal muscle relaxants (SMRs)

① Local Anaesthetics → Sodium blockers

→ Drugs which on topical application or local injection causes reversible loss of pain & sensation.



MOA → Block nerve conduction by decreasing the entry of Na^+ ions during upstroke of action potential.

② SMRs → Acts by blocking depolarization

Drugs → ① Succinyl choline

② Curares → Pancuronium, Doxacurium, Vencuronium

Antidote → Neostigmine

③ Autacoids & Related Drugs

Autacoids → Produced by wide variety of cells in the body, having intense biological activity, but generally act locally at the site of synthesis & release (Local hormones)
e.g → Histamine, serotonin, prostaglandin, etc.

NSAIDs → Non Steroidal Anti Inflammatory Drugs.

→ All NSAIDs have analgesic, antipyretic & antiinflammatory actions.

MOA → NSAID's act by inhibiting the PG synthesis (Proteoglandin)
→ Prostaglandin gives protection to gastric mucosa against HCl.

Group-I → Non-selective COX (cyclo-oxygenase) inhibitors (i.e. inhibits both COX-I & II)

- | | |
|--------------------------------|--------------------|
| 1) Aspirin | 6) Indomethacin |
| 2) Paracetamol (Acetaminophen) | 7) Mefenamic acid |
| 3) Ibuprofen | 8) Ketorolac |
| 4) Diclofenac sodium | 9) Phenyl butazone |
| 5) Piroxicam | |

Group-II → Selective NSAIDs (i.e. only inhibits COX-II) or Stomach friendly

- 1) Nimesulide
- 2) Meloxicam
- 3) Etoricoxib

② SMRs → Acts by blocking depolarization

Drugs → ① Succinyl choline
② Curares → Pancuronium, Doxacurium, Vencuronium

Antidote → Neostigmine

2) Antipyretics

MOA → Acts by inhibiting the enzyme cyclo-oxygenase, reducing the level of Prostaglandins at the hypothalamus.

Drugs → 1) Aspirin 2) Acetaminophen (PCM)
 3) Ibuprofen 4) Diclofenac sodium

3) Analgesics

1) Aspirin 2) PCM 3) Ibuprofen 4) Diclofenac sodium

④ Drugs for Respiratory Disorders

① Bronchodilators

→ Substances that dilates bronchi & bronchioles decreasing resistance in the respiratory airway & increased airflow to lungs.

Classifⁿ →

β_2 - Sympathomimetics

Short Acting

- i) Salbutamol
- ii) Terbutaline

Long Acting

- i) Salmeterol
- ii) Formoterol
- iii) Bambuterol

Methylxanthiness

- i) Theophylline
- ii) Aminophylline
- iii) Doxophylline.

MOA → Highly selective β_2 agonist causes bronchodilation through β_2 receptors. Stimulation by increasing CAMP (cyclic AMP) formation in bronchial muscle cells resulting in relaxation of bronchial smooth muscle

② Inhalants

MOA → They enter bloodstream quickly & distributed throughout brain & body. Hence they have direct effect on CNS & PNS

Classifⁿ

- ① Asphyxiants → Helium
- ② Depressants → Gasoline
- ③ Anaesthetics → Nitrous oxide.

③ Expectorants

These drugs increase bronchial secretion, reduce its viscosity facilitating its removal by coughing

→ Sodium & Potassium citrate

1) Antihistamines drugs

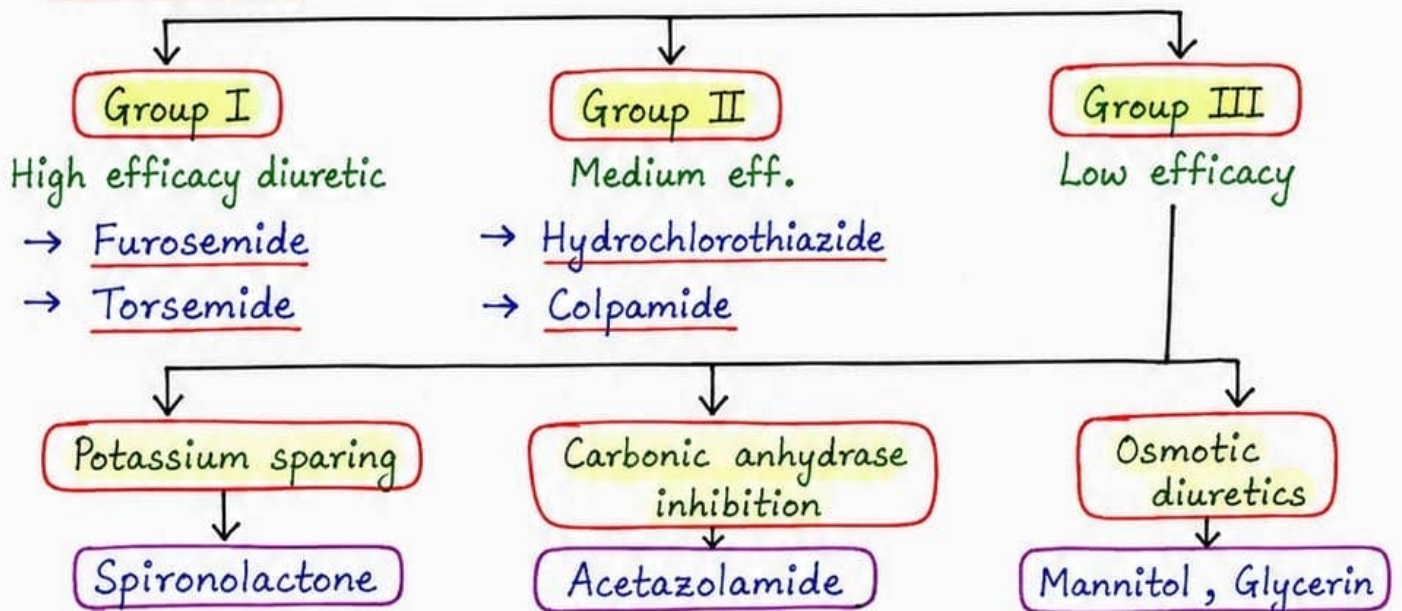
These drugs act in use to relief the threshold of cough centre or act peripherally in respiratory tract to reduce tussal impulse or both these action.

- Examples –
- 1) Codeine
 - 2) Noscapine (Narcotine)
 - 3) Dextromethorphan

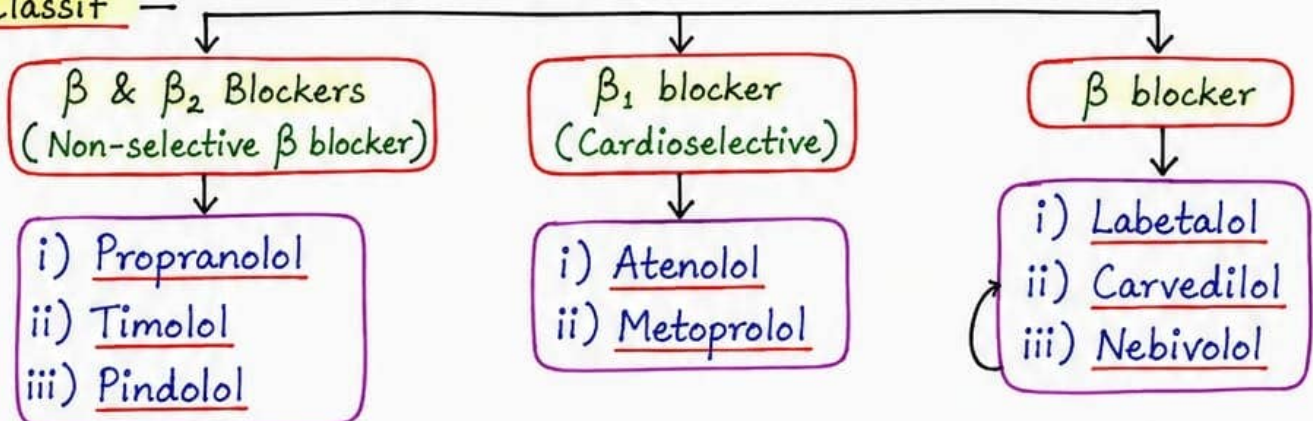
Cardiovascular Drugs (3)① Antihypertensives → ⑤

(A) Diuretics → Diuretics are drugs which increase urine output by causing net loss of Sodium & water in urine.

MOA → It inhibits Na^+ , K^+ , Cl^- & cotransport by acting on thick ascending loop of Henle's.

Classification —(B) β -Blockers

MOA → These drugs blocks adrenergic response mediated through β -receptors.

Classifⁿ —

(C) Calcium channel Blockers (C.C.B)

MOA → They block voltage sensitive calcium channels hence called as C.C.B.

Classifⁿ →



(D) ACE Inhibitors Angiotensin converting Enzyme inhibitors

MOA → Decreases BP by inhibiting enzyme that convert angiotensin I to angiotensin II.

Drugs → 1) Captopril
2) Enalapril
3) Ramipril

(E) Angiotensin II receptor antagonist

MOA → It blocks all action of angiotensin II, i.e. it blocks vasoconstriction sympathetic stimulation & release of adrenaline & aldosterone.

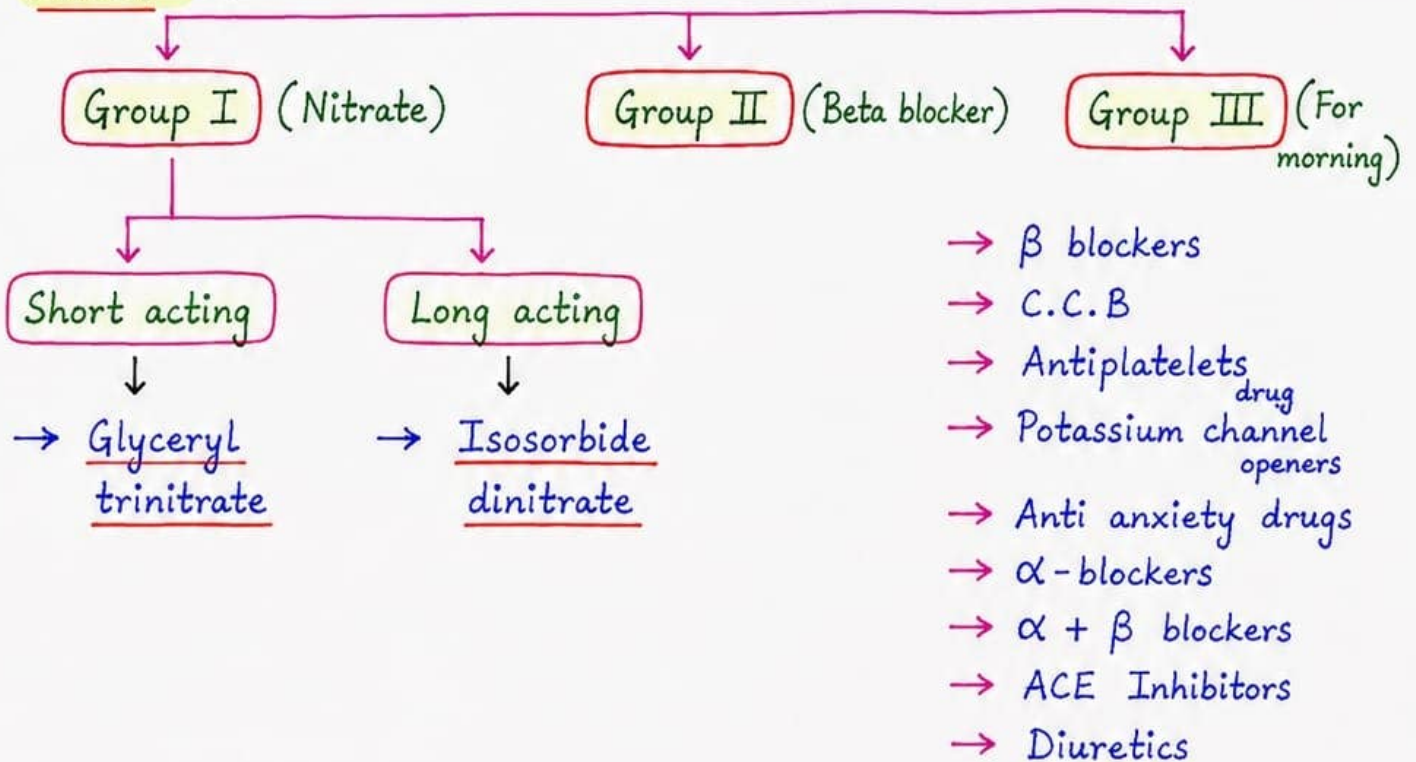
Drugs → 1) Losartan
2) Telmisartan
3) Olmisartan

1 Antiinanginal drugs (Metha classification)

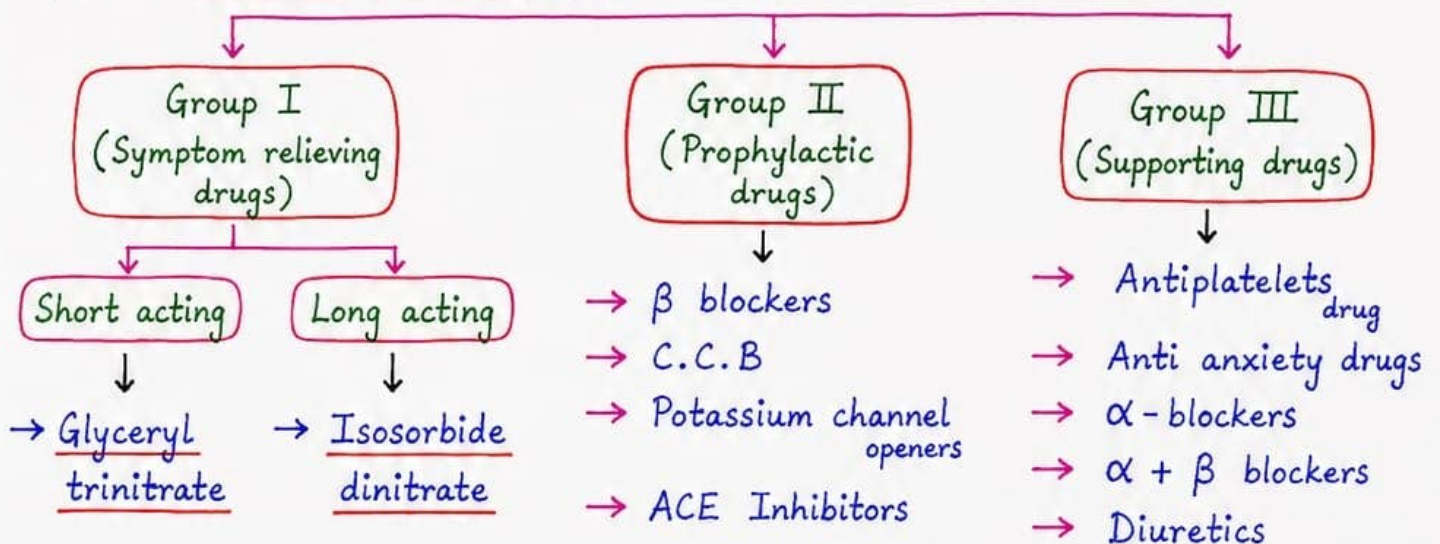
Angina Pectoris is a pain syndrome due to induction of an adverse oxygen supply situation in a portion of myocardium.

→ Angina pectoris means inadequate O₂ supply to heart.

Classifⁿ →



2 Metha classification



Drug Affecting Blood

① Haematinics

MOA → Play source, supplementation role in haematopoiesis

- Drugs →
- 1) Iron preparations ✓
 - 2) Cobalamine ✓
 - 3) Erythropoietin ✓

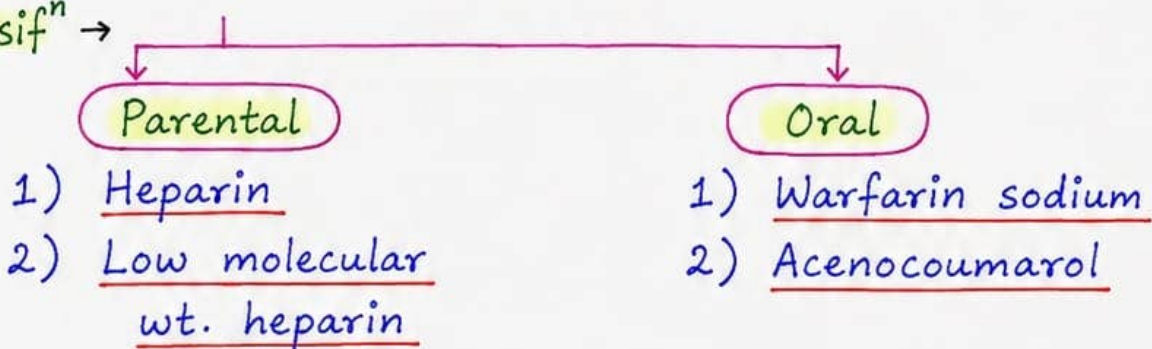
② Coagulants

- Drugs →
- 1) Vit. K ✓
 - 2) Fibrinogen ✓
 - 3) Desmopressin ✓

③ Anticoagulants

MOA → Heparin combines with plasma antithrombin III & binds with clotting factor's of intrinsic & ~~z~~ common pathways, i.e. factor ii, X, IX

Classifⁿ →



④ Hypolipidaemic drugs

- | | |
|--------------------------|--|
| 1) <u>Nicotinic acid</u> | 3) <u>Neomycin</u> |
| 2) <u>Guggu lipid</u> | 4) <u>Bile acid active draining agents</u> |

Gastrointestinal drugs (2)

- ① Antacids → These are the basic substances which neutralize gastric acid & raise pH of gastric contents.
→ They lost popularity because they only neutralize & not decreases HCl secretion.

MOA → ① Neutralize the acid & at low doses promote gastric mucosal defensive mechanism.

- ① Irreversibly block the hydrogen or potassium adenosine triphosphatase enzyme, thereby parietal cell can't produce HCl.

Classifⁿ →

Systemic

- Sodium bicarbonate
- Sodium citrate
- Aluminium hydroxide gel

Asystemic

- Magnesium hydroxide

- ② Proton pump (Inhibitors): most efficacious inhibitors of the gastric acid secretion.

- 1) Omeprazole
- 2) Pantoprazole

H₂ receptor antagonist →

- 1) Ranitidine
- 2) Famotidine

- ② (Carminatives) These drugs which promote the expulsion of gases from G.I.T & give a feeling of warmth & comfort in epigastrium.

e.g → Peppermint oil, tincture ginger.

- Drugs → 1) Simethicone
2) Dimethicone

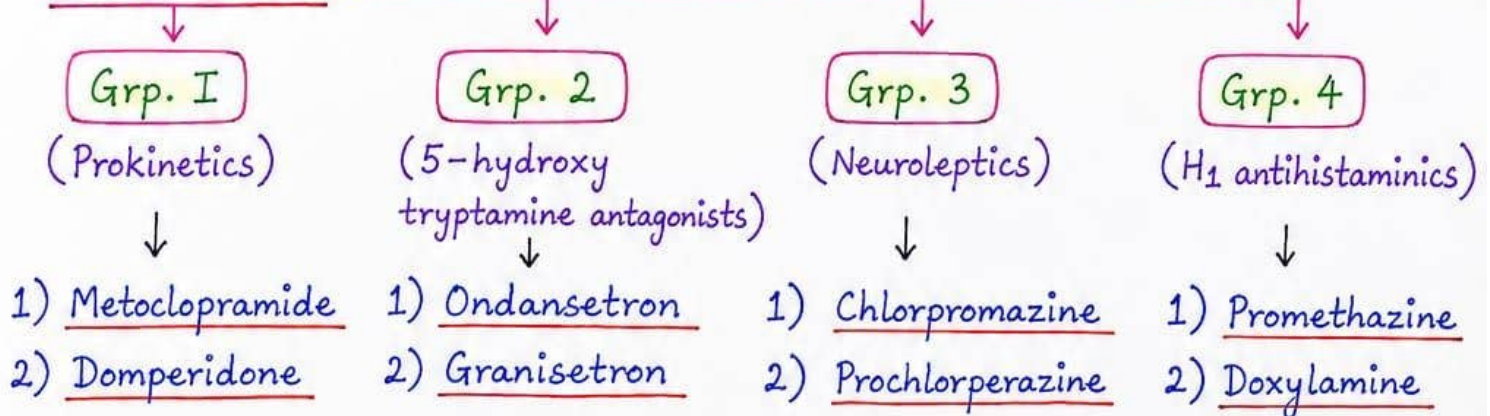
B. Digestants → These substances which promotes the digestion of food.

- 1) HCL
- 2) Pepsin
- 3) Papain
- 4) Diastase
- 5) Pancreatin → (Proteolytic enzyme from papaya)

C. Antiemetic drugs

- Effective against nausea & vomiting
- Typically used to treat motion sickness, side effects of opioid analgesics, general anaesthesia.

Classification :



5. Laxatives

- ① Bulk forming → Methyl cellulose
- ② Stool softener → Liquid paraffin
- ③ Stimulant purgative → Bisacodyl, castor oil
- ④ Osmotic purgative → Magnesium sulphate (Mannitol)



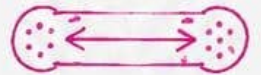
6. Antidiarrhoeal drugs

- ① IV fluids ORS
- ② Ciprofloxacin & Norfloxacin
- ③ Metronidazole , Tinidazole , • Ornidazole
Tab. Lometil

7. Hepatoprotective drugs

- 1) Ursodeoxycholic acid
- 2) Penicillamine

Antibacterial Drugs



Antibiotics → Antibiotics are substances produced by micro-organisms which ~~kill~~ suppress the growth of the micro-organisms, i.e. bacteriostatic or kill the micro-organisms without affecting the host, i.e. bactericidal.

- Narrow spectrum → Effective only either against gram +ve or gram -ve.
- Broad Spectrum → Effective against both.

Groups

- 1) β -lactam antibiotics
- 2) Fluroquinolones
- 3) Sulfonamide
- 4) Cotrimoxazole
- 5) Tetracycline
- 6) Macrolides
- 7) Amino glycosides

BFS (TMA)

①. β -lactam antibiotics → Having β -lactam ring.

→ β -lactam antibiotics may not be effective against organism producing β -lactamase or penicillin's therefore nowadays they are combined with β -lactamase inhibitor to increase the antibacterial activity of Penicillin's & Cephalosporin's.

β -lactamase inhibitors →

- 1) Clavulanic acid
- 2) Sulbactam
- 3) Tazobactam
- 4) Amibactam

Drugs: (A) Penicillins → King of antibiotic

MOA → Bactericidal antibiotic which acts by interfering with synthesis of bacterial cell wall.

(Uses) Used in streptococcal infections like pharyngitis, otitis media in haemophilus influenzae.

- 1) Pneumococcal infections
- 2) Gonorrhoea
- 3) Syphilis

Penicillins used duration of action (penicillin class)

- | | |
|--------------------------------|----------------------------|
| 1) Short acting penicillin | } (Albertstein sumukheess) |
| 2) Intermediate acting | |
| 3) Long acting (Inj. Penidura) | |

(B) Semisynthetic Penicillin → 3 types

MOA
(same)

(I) Broad spectrum semisynthetic penicillin

- Ampicillin
- Amoxicillin
- Piperacillin

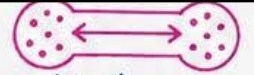
(II) Penicillinase - resistant Penicillin's

- cloxacillin

(III) Acid - resistant alternative to Penicillin G

(C) Cephalosporins (Same MOA)

- A) First generation → cefazolin
- B) Second generation → cefuroxime
- C) Third generation → cefotaxime, ceftriaxone
- D) Fourth generation → cefepime



① Monobactams → Aztreonam → used in hospital acquired infection (Nosocomial infection).

② Carbapenems → 1) Meropenem
2) Imipenem

→ Extremely potent & used in resistant cases after culture & sensitivity report.

③ Fluoroquinolones (FQS)

MOA → FQs are bactericidal & act by inhibiting enzyme bacterial DNA gyrase.

A) First Generation FQS → 1) Norfloxacin
2) Ofloxacin

B) Second Generation FQS → 1) Levofloxacin
2) Moxifloxacin
3) Lomefloxacin

④ Sulfonamides → Effective against pyogenic bacterial infections (Pus producing bacteria)

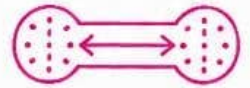
Group I → Sulfadiazine

Group II → Sulfamethoxazole

Group III → Sulfamethoxypyridazine

Group IV → Sulfacetamide (used in conjunctivitis)

MOA → Inhibit synthesis of folic acid.



- ①. Cotrimoxazole → combination of
Trimethoprim 80 mg + Sulfamethoxazole
400 mg

MOA → Acts by producing sequential block in Folic acid synthesis

→ Chloramphenicol (use -- Eye infections - Aplcays
(तसालीन-use))

- ⑤. Tetracyclines

MOA → Bacteriostatic, & inhibit protein synthesis by
binding to 30S ribosome in susceptible organisms.

- 1) Tetracycline & oxytetracycline
- 2) Doxycycline

- ⑥. Macrolides → Bacteriostatic in low doses &
bactericidal in high doses.

MOA → Act by inhibiting bacterial protein synthesis.

Use → Popularly used in RTI.

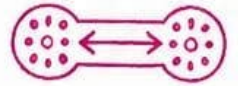
Drugs → 1) Erythromycin
2) Azithromycin

- ⑦. Aminoglycosides

MOA → Bactericidal & act by disturbing the protein synthesis
so that bacterial cell membrane is not well formed.

Drugs → 1) Streptomycin 3) Amikacin 5) Neomycin
2) Gentamycin 4) Tobramycin

Tuberculosis is a chronic granulomatous inflammatory disease of lung parenchyma caused by acid-fast bacilli — Mycobacterium tuberculosis.



Anti-tubercular drugs

MOA - Tuberculocidal & acts by inhibiting synthesis of mycolic acid which is essential component of cell wall.

- ① Acts by inhibiting DNA dependent RNA synthesis.
- ② Kills intracellular bacilli.

Classification

(1) First line drugs

- | | | | |
|-----------------|-----|---|------------------|
| 1) Isoniazid | (H) | } | Oral route |
| 2) Rifampicin | (R) | | |
| 3) Pyrazinamide | (Z) | | |
| 4) Ethambutol | (E) | | |
| 5) Streptomycin | (S) | → | Injectable route |

(2) Second line drugs

- 1) Thiacetazone (Tzn)
- 2) Para-aminosalicylic acid (PAS)
- 3) Ethionamide (Etn)
- 4) Cycloserine (Cys)
- ✓ 5) Amikacin (Am)
- ✓ 6) Macrolides
- ✓ 7) Fluroquinolones

(3) Sulfonamides → Effective against pyogenic bacterial infections (Pus producing bacteria)

Group I → Sulfadiazine

Group II → Sulfamethoxazole

Group III → Sulfamethoxypyridazine

Group IV → Sulfacetamide (used in conjunctivitis)

MOA → Inhibit synthesis of folic acid.

- Fungal infection caused by dermatophytes & candida.
- Fungal infection common in → Diabetes mellitus
 - Unhygienic & moisty in summer due to excessive sweating.

Antifungal

Antifungal

Group-I → Antibiotics → Amphotericin
Heterocyclic benzothyan

Group-II → Azoles → (A) Imidazoles

- Clotrimazole
- Ketoconazole

(B) Triazoles

- Fluconazole
- Itraconazole

Group-III → Allylamine → Terbinafine

Antiviral

MOA → Inhibit DNA synthesis so that there is no viral replication.

1) Group-I — Antiviral drugs against Herpes virus
namely — Acyclovir

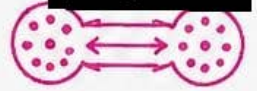
2) Group-II — Anti-retrovirus (Drugs used in AIDS)

- 1) Stavudine 2) Lamivudine
- 3) Nevirapine 4) Indinavir

3) Group-III — Anti-influenza drugs

- 1) Amantadine

→ Malaria is a parasitic infⁿ caused by species of Plasmodium.



Antimalarial

MOA → Kills schizonts by increasing pH of erythrocytes so that there is accumulation of toxic substance in erythrocytes which kills schizonts.

Drugs →

- 1) Chloroquine
- 2) Quinine
- 3) Mefloquine
- 4) Artesunate
- 5) Artemether

Antihelminthic Drugs

MOA → Blocks total or partial glucose uptake in parasite so that parasite get paralysed & expelled out.

Drugs →

- 1) Mebendazole
- 2) Albendazole
- 3) Ivermectin.

Hormones & Related Drugs

- 1) Thyroid hormones → Thyroxine (T_4)
→ Tri-iodothyroxine (T_3)
→ Calcitonin.

2) Thyroid Inhibitor (Antithyroid)

MOA → Inhibit release of thyroid hormones.

Drugs → 1) Carbimazole
2) Lugol's iodine
3) Radioactive iodine

Insulin → Insulin is a normal physiological hormone secreted by β -cells of pancreatic islets of Langerhans to decrease blood glucose.

Normal BSL → 80 - 120 mg/dl

← Fasting
(In morning hrs)
- empty stomach

Post prandial
↳ 2 hrs of lunch
P.P → 80 - 160
(after 2 hrs of lunch)

MOA → Insulin ↓ blood glucose levels by acting on -
a) Liver b) muscles c) Adipose tissue.

Prepⁿ of Insulin

(A) Conventional or older prepⁿ of → Produced from beef & pork pancreas & they are called as 'Bovine insulin' & 'porcine Insulin'.

(B) Newer insulin prepⁿ of → ① Highly purified insulin.
② Human insulin
↳ By rDNA technique.

Hormonal contraceptive

- ① Injectable — Depomedroxy progesterone acetate
- ② Subcutaneous implants
- ③ Vaginal ring
- ④ Oral pills — Progesterone & combined pill

Uterine stimulants (Oxytocin drugs)

- 1) Serotonergic agents
- 2) Ergonovine
- 3) Prostaglandins

Uterine relaxants

- 1) Adrenergic agonists — Ritodrine
- 2) Calcium channel blockers
- 3) Mg. sulfate
- 4) Oxytocin antagonists

Antiseptic drugs

- 1) Phenol derivatives → phenol
- 2) Oxidizing agents → Hydrogen peroxide
- 3) Halogens → Iodine
- 4) Alcohols → Ethyl alcohol
- 5) Aldehydes → Formaldehyde

Disinfectants

- 1) Inorganic → Halogen, oxidizing agents, metallic salts
 - 2) Organic → Aldehyde, phenol, dyes, detergents
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